

F : field

V : vector space over F

Defⁿ

$X \subseteq V$: subset

The span of X written

$\text{span}(X)$ is the smallest subspace of V containing X

Rmk (1) By definition

$\text{span}(X) \subseteq V$

is a subspace

$$(2) \text{ If } X = \emptyset$$

then

$$\text{Span}(X) = \{0_V\} \subseteq V$$

Note: $\{0_V\} \subseteq V$ is a
subspace!

$$(3) \quad 0 \neq v \in V$$

$$X = \{v\}$$

$$X \subseteq \text{Span}(X) = \underbrace{\{a \cdot v : a \in F\}}_{\substack{\downarrow \\ V}}$$

(1) (closure under $+$)

$$a_1 \cdot v + a_2 \cdot v = (a_1 + a_2) \cdot v$$

(2) (closure under $\cdot$)

$$a_1 \cdot (a_2 \cdot v) = (a_1 a_2) \cdot v$$

Observation:

$\text{Span}(X)$ is characterized by:

If $W \subseteq V$ is any subspace with $X \subseteq W$, then

$$\text{Span}(X) \subseteq W.$$

$$(4) \quad X = \{v_1, v_2, \dots, v_m\}$$

$$v_1, v_2, \dots, v_m \in V$$

Then

$$\text{Span}(X) = \left\{ a_1 v_1 + a_2 v_2 + \dots + a_m v_m : \right. \\ \left. a_i \in F \right\}$$

Defⁿ Given $v_1, \dots, v_m \in V$, an

F -linear combination (or just linear combination)

of $v_1, \dots, v_m$ is a vector
of the form

$$a_1 v_1 + a_2 v_2 + \dots + a_m v_m = \sum_{i=1}^m a_i v_i$$

for $a_1, a_2, \dots, a_m \in F$

So

$\text{Span}\{v_1, v_2, \dots, v_m\}$

$= \left\{ \begin{array}{l} \text{Linear combinations} \\ \text{of } v_1, v_2, \dots, v_m \end{array} \right\}$

Examples

$$\textcircled{1} \quad \text{Poly}(F)$$

$$\begin{array}{c} \cup \\ \{1, x, x^2\} \end{array} \subsetneq X$$

$$\text{Span}(\{1, x, x^2\}) = \text{Poly}_{\leq 2}(F)$$

$$\begin{array}{c} \cup \\ \{a_1 \cdot 1 + a_2 \cdot x + a_3 \cdot x^2 : a_1, a_2, a_3 \in F\} \end{array}$$

More generally,

$$\text{Span}(\{1, x, \dots, x^m\}) = \text{Poly}_{\leq m}(F)$$

②

$$n, m \in \mathbb{N}$$

plural of
'matrix'

$$M_{n \times m}(F) = \left\{ \begin{array}{l} n \times m - \text{matrices} \\ \text{with coefficients} \\ \text{in } F \end{array} \right\}$$

An $n \times m$ -matrix over F
is an array of the
form

$$\begin{bmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,m} \\ a_{2,1} & \cdots & \cdots & a_{2,m} \\ \vdots & & \ddots & \vdots \\ a_{n,1} & a_{n,2} & \cdots & a_{n,m} \end{bmatrix}$$

$$a_{i,j} \in F,$$

$$\begin{array}{l} 1 \leq i \leq n \\ 1 \leq j \leq m \end{array}$$

Ex 2

$$M_{1 \times 1}(F) = \{ [a] : a \in F \}$$

$$M_{n \times 1}(F) = \left\{ \begin{bmatrix} a_{1,1} \\ a_{2,1} \\ \vdots \\ a_{n,1} \end{bmatrix} : a_{k,1} \in F \right\}$$

$$= F^n$$

$M_{n \times m}(F)$ is an F -vector space

$$(1) \begin{bmatrix} a_{1,1} & \dots & a_{1,m} \\ \vdots & & \vdots \\ a_{n,1} & \dots & a_{n,m} \end{bmatrix} + \begin{bmatrix} b_{1,1} & \dots & b_{1,m} \\ \vdots & & \vdots \\ b_{n,1} & \dots & b_{n,m} \end{bmatrix}$$

$$= \begin{bmatrix} a_{11} + b_{11} & \dots & a_{1m} + b_{1m} \\ \vdots & & \vdots \\ a_{n1} + b_{n1} & \dots & a_{nm} + b_{nm} \end{bmatrix}$$

(2) CEF

$$C \cdot \begin{bmatrix} a_{11} & \dots & a_{1n} \\ \vdots & & \vdots \\ a_{n1} & \dots & a_{nm} \end{bmatrix}$$

$$= \begin{bmatrix} C \cdot a_{11} & \dots & C \cdot a_{1n} \\ \vdots & & \vdots \\ C \cdot a_{n1} & \dots & C \cdot a_{nm} \end{bmatrix}$$

$$X = \left\{ \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \right\}$$

$$\subseteq M_{2 \times 2}(F)$$

$$\text{Span}(X)$$

$$= \left\{ a_1 \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + a_2 \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \mid a_1, a_2 \in F \right\}$$

$$= \left\{ \begin{bmatrix} 0 & a_1 \\ a_2 & 0 \end{bmatrix} : a_1, a_2 \in F \right\}$$

anti-diagonal

diagonal

Q: Suppose we have

$$X \subseteq V, \text{ and } v \in V$$

How can we tell if

$$v \in \text{Span}(X)??$$

Example: $F = \mathbb{R}$

$$V = \mathbb{R}^3$$

$$X = \left\{ \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} \right\} \subseteq \mathbb{R}^3$$

$$v = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} \in \mathbb{R}^3$$

Then the question is:

$$\text{Is } v = a_1 \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} + a_2 \begin{bmatrix} \pi \\ -1 \\ 2 \end{bmatrix}$$

for $a_1, a_2 \in \mathbb{R}$?

i.e. Can we find $a_1, a_2 \in \mathbb{R}$

s.t.

$$\begin{bmatrix} a_1 + \pi a_2 \\ a_1 - a_2 \\ 2a_1 + 2a_2 \end{bmatrix} \stackrel{??}{=} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}$$